Work Experience Survey

For students in Earth and Environmental Sciences

Have you: Applied for a work experience placement?
Done work experience?
Never considered doing work experience?

If so, we want to hear from you! Please complete this very short survey about work placements.
This is to help us review how we support work placements and to support our accreditation cases

Deadline 21st November

https://forms.office.com/e/fn0rYpL21x

EART26201 part 2: fluid geochemistry

Course syllabus:

1. Water as a solvent - ions
2. Solubility and complexing
3. Acids and bases
4. Redox reactions
5. Stable isotopes in the hydrologic cycle

Unit: EART26201

Department: Department of Earth and Environmental Sciences

Weeks: 2-13 (29 Sep 2025-21 Dec 2025)

	9:00	9:30	10:00	10:30	11:00	11:30	12:00	12:30	13:00	13:30	14:00	14:30	15:00	15:30	16:00	16:30	17:00	17:30
Mon									EART26201/LECTURE1/ Stopford_TH 4 Neave, David;Holland, Greg LECTURE 2-6, 8-12									
Tue																		
Wed	EART26201/LABORATORY/02 Williamson_1.18 Gaskell, Jordan;Holland, Greg LABORATORY 8, 10																	
	EART26201/WORKSHOP/ Williamson_G.12/G.14 Neave, David;Holland, Greg WORKSHOP 2-6, 9, 11-12																	
Thu	EART26201/LABORATORY/01 Williamson_1.18 Holland, Greg;Gaskell, Jordan LABORATORY 8, 10						EART26201/LABORATORY/03 Williamson_1.18 Holland, Greg;Gaskell, Jordan LABORATORY 8, 10											
Fri	EART26201/LABORATORY/04 Williamson_1.18 Holland, Greg;Gaskell, Jordan LABORATORY 8, 10																	

Previous Week
Back To Selection

Next Week

Thu Jul 24 2025 10:40:12 GMT+0100 (British Summer Time)

Basic revision: structures of atoms

Reference: ^{12}C = 12.00000 atomic mass units (amu)
1 mole = 6.02252 x 10^{23} atoms = Avogadro's number

Therefore 1 amu = 1/12 x 12.00000 / 6.02252 x 10^{23} = 1 66043 x 10^{-24} g

Particle	Symbol	Charge (C)	Mass (amu)
Proton[1]	Z	^{+}e = 1.6021 x 10^{-19}	1.007825
Neutron[2]	N	zero	1.008665
Electron	e	^{-}e = 1.6021 x 10^{-19}	0.0005486

[1]Atomic number
[2]Atomic mass (A) = Z + N (amu)

Notice that mass of ^{12}C = 6N + 6Z + 6e = 12.098931
ΔM= sum of masses of protons and neutrons - measured mass of nucleus
= 12.098931-12.00000- = 0.098931 amu
It represents the binding energy of the nucleus ($E=mc^2$)

Basic revision: Stoichiometry

A **mole** of a substance equals the **gram-formula mass**

e.g. NaCl, atomic mass Na = 22.9 atomic mass; Cl = 35.45 amu

molecular mass NaCl = 22.9+35.45 = 58.35 amu

formula mass = 58.35 amu

gram-formula mass = 58.35 grams

Molarity (M): the number of moles of a solute dissolved in a litre of solution.

A molar solution is made by placing 1 mole of a solute into a 1 litre volumetric flask. Water is then added to the volumetric flask up to the one litre mark.

For a 1 molar solution of NaCl we would weigh out 58.35 g

Basic revision: Stoichiometry

Molality (m) : Molality is the number of moles of solute dissolved in one kilogram of solvent.

Be careful not to confuse molality and molarity. Note that the solvent must be weighed. Because one litre of water has a specific gravity very close to 1.0 at room temperature and weighs nearly one kilogram; one can measure out one litre of water and add the solute to it.

When the solvent is water molarity and molality are very similar.

To make a one molal aqueous (water) solution of sodium chloride (NaCl), measure out one kilogram of water and add one mole (58.35g) of NaCl to it.

Water as a Solvent

Water - most abundant liquid on Earth

- key role in most geochemical and biological processes

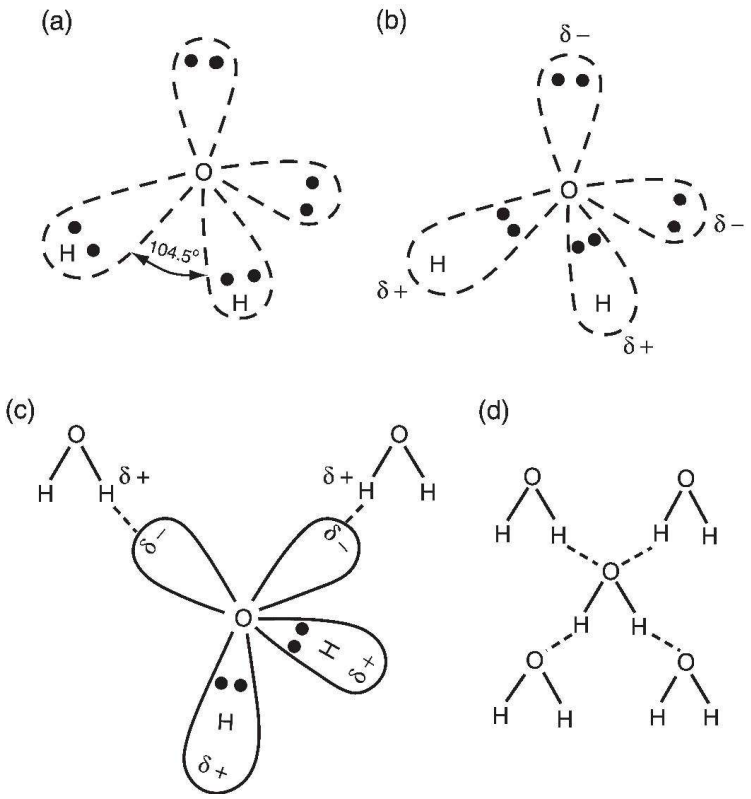
- the medium in which chemical reactions take place in nature

Special properties of Water

- Higher boiling point than other hydrides of higher molecular weight (e g H_2S)
  - liquid at most surface conditions
- Very high heat capacity relative to common liquids and solids
  - means oceans are an important heat sink for the Earth's climate
- Maximum density of water is at 4 deg-C
  - means ice floats on the surface and does not sink to the bottom of the ocean – **what would happen if it did?**
- An excellent solvent for ionic compounds i.e. compounds that dissociate into negatively and positively charged ionic species in water.

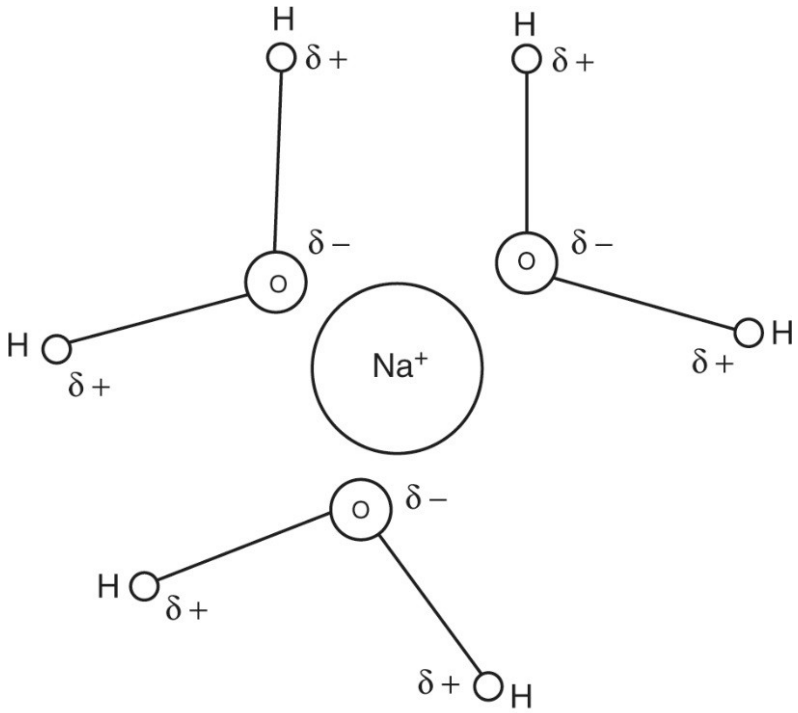
Water as a Solvent - Ions

- Pure water is a strong dielectric = an electrical insulator
- The high dielectric constant of water is a measure of its polarity and is responsible for the remarkable solvent properties of water.
- Solute ions are hydrated by the dipolar water molecules preventing them from re-combining.



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Electrolyte solutions



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When NaCl dissolves in water it dissociates into ions:

$$NaCl_{(s)} = Na^{+} + Cl^{-}$$

Results in hydrated cations and anions

High capacity of water to dissolve ionic compounds means:

- Water in nature almost always contains dissolved ionic species (an aqueous solution)
- Dissolved ions are charged particles or electrolytes (conduct electricity) so aqueous solutions are electrolyte solutions

Activity Coefficient

- In very dilute aqueous solutions, ions do not interact with each other.
- This is the standard state (ideal solution) for ions in aqueous solution.
- With increasing concentration, ions start to interact with each other (non-ideal solutions). The chemical potential of an ion will depend upon the concentration of the other ions.
- Ion-ion interactions may be specific (e.g. ion pairs or complexes) or non-specific (long-range coulombic interactions).
- The net results is that chemical reactions among ions and molecules in aqueous solution are often not a direct function of just their concentrations.

Ion speciation in seawater

Species	Molality	Species	Molality
Na^+	4.8×10^{-1}	Ca^{2+}	9.6×10^{-3}
$NaSO_4^-$	6.0×10^{-3}	$CaSO_4$	1.1×10^{-3}
$NaHCO_3$	1.7×10^{-4}	$CaHCO_3^-$	4.6×10^{-5}
K^+	1.0×10^{-2}	Mg^{2+}	4.8×10^{-2}
Cl^-	5.7×10^{-1}	$MgSO_4$	7.4×10^{-3}
SO_4^{2-}	1.5×10^{-2}	$MgHCO_3^-$	2.2×10^{-4}
HCO_3^-	1.5×10^{-3}		

Activity Coefficient

- We therefore have to replace concentration with ***activity***.
- The activity ***a*** of an ion in solution is related to its molal concentration ***m*** by the activity coefficient γ such that :

$$a = \gamma m$$

- Activities are expressed as molalities (number moles solute per kg of water)
- The activity coefficients (γ) correct (effectively reduce) the concentrations of ions resulting from the interferences by other ions in solutions.
- Values of γ are <1 which means the effective concentration or activity of ions is ***less*** than their actual concentration.
- The difference between activity and concentration is unimportant in very dilute solutions (ideal solutions) i.e. activity = concentration.
- Activity coefficients of individual ions in solution are determined using the Debye-Hukel theory and extensions (see Misra 7.2.3 for details).

Ionic strength

- The concentrations and charges of ions in a solution are expressed by its ionic strength **I**.

$$I = \frac{1}{2}\sum m_i z_i^2$$

- Where m_i are concentrations expressed in moles and z_i are the charges on the ions.
- The ionic strength of most natural waters is less than 0.1 mol/Kg.
- Seawater has an ionic strength of 0.7 mol/Kg and oilfield brines can be 5x higher than this.

	Ganges River water[a]		Groundwater, limestone aquifer, Florida[b]		Surface seawater[c]	
Ions	mg L^{-1}	Molality ($\times 10^{-3}$)†	mg L^{-1}	Molality ($\times 10^{-3}$)†	mg L^{-1}	Molality ($\times 10^{-3}$)†
Ca^{2+}	25.4	0.634	56	1.397	406	10.13
Mg^{2+}	6.9	0.284	12	0.494	1271	52.29
Na^{+}	10.1	0.44	7.9	0.34	10570	460
K^{+}	2.7	0.069	1.0	0.026	380	9.719
Cl^{-}	5	0.14	12	0.34	19011	536
Br^{-}	–	–	–	–	66	0.826
SO_4^{2-}	8.5	0.088	53	0.552	2664	27.73
HCO_3^{-}	127	2.082	160	2.623	121	1.983
CO_3^{2-}	–	–	–	–	18	0.3
I (mol kg^{-1} water)		0.0034		0.0066		0.6852

[a]Sarin *et al.* (1989); [b]Goff (1971) as quoted in McSween *et al.* (2003); [c]Berner (1971).
*The concentration shown for any ionic species represents the sum of free ions plus ion pairs formed with ions of opposite charge. For example, the concentration of SO_4^{2-} in seawater is the sum of concentrations of the following species: SO_4^{2-}, $NaSO_4^{-}$, $CaSO_4^{0}$, and $MgSO_4^{0}$.
†To (simplify calculations it is assumed here that 1 L of water weighs 1 kg.

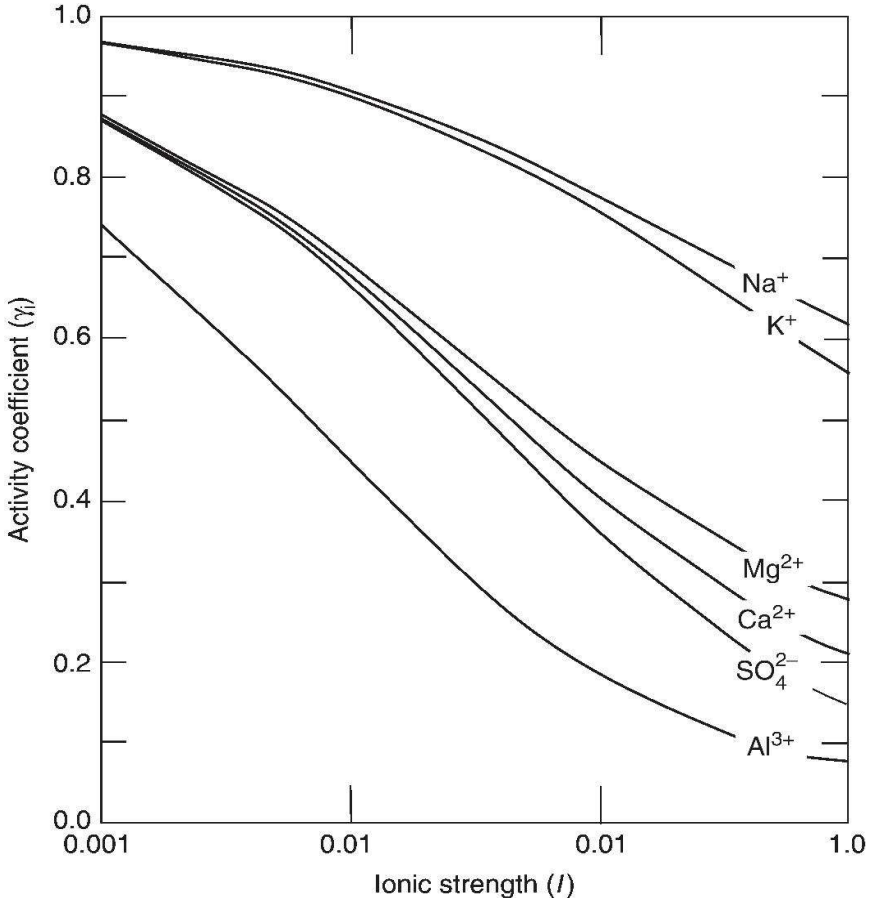
Table 7.1 Analysis and calculated ionic strengths of some natural waters*.

Note that H^+ and OH^- are not included because in neutral solutions their concentrations are very low (10^{-7} mol/kg)

Ca^{2+} molality calculation for Ganges:
grams of Ca^{2+} = 25.4/1000 = 0.0254 g/L
(electrons have no mass)

moles Ca^{2+} = 0.0254/40.08 = 0.000634
=0.634 × 10^{-3} mol/kg

Relationship between ionic strength (I) and activity (γ)



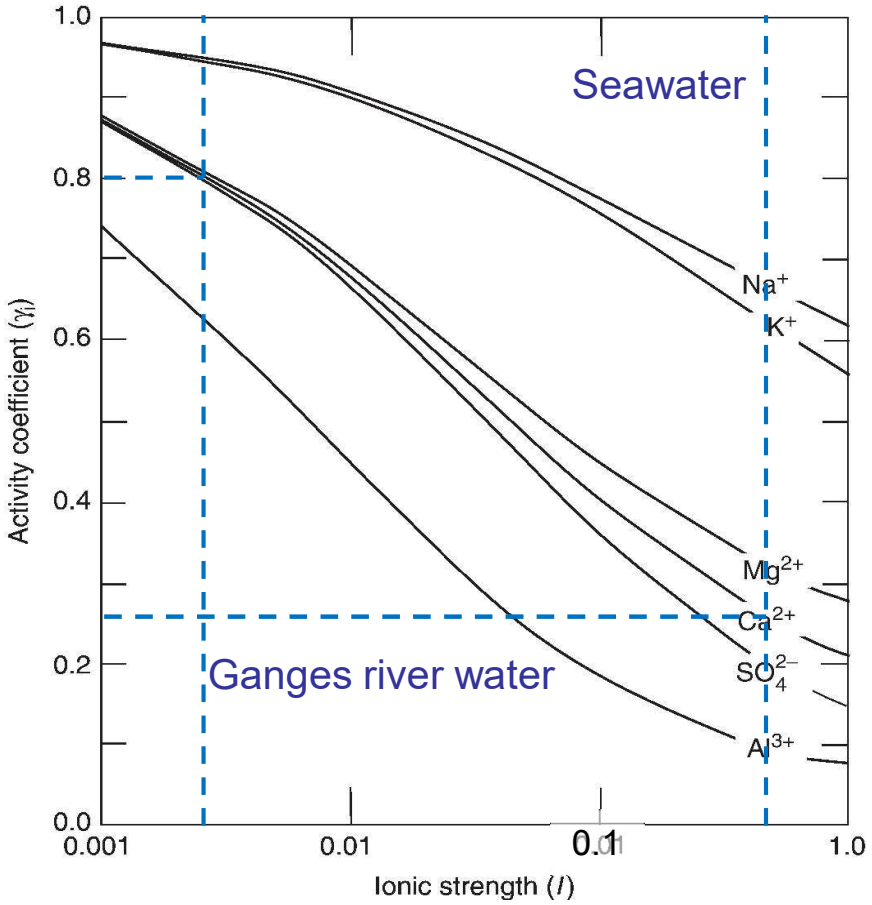
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Fig. 7.3 Activity coefficients of some common aqueous species at 298.15 K (25°C) and 1 bar as a function of ionic strength, calculated using the extended Debye–Hückel equation (equation 7.6) and values of the parameters given in Table 7.2. Note that for a particular ion, the activity coefficient decreases with increasing ionic strength and for ions of equal charge the differences in activity coefficients due to different values of the *å* parameter become important at $I > 0.01\,m$.

Key points

- γ decrease as charge of ion increases.
- γ decrease with ionic strength (I)
- This can be understood as ion “interference” as the ionic strength increases.

Relationship between ionic strength (I) and activity (γ)



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Ganges river water (I = 0.0034) the activity of Ca^{2+} is 0.80 and in seawater (I = 0.685) it is about 0.26.

This means the presence of other ions reduces the effective concentration of Ca^{2+} by 20% and 74% in Ganges river water and seawater respectively.

Equilibrium constant

The Law of Mass Action

Reactions reach a state of equilibrium when the rate of forward reaction equals the rate of backward reaction

$$A + B \leftrightarrow C + D$$

a, b, c, d are molar coefficients taken from the balanced equation for the reaction
[A], [B], [C] and [D] are concentrations of the reactants and products **at equilibrium**

$$a\,A + b\,B \leftrightarrow c\,C + d\,D$$

K is the ***equilibrium constant*** of the reaction, in general: $K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$

e.g.

$$H_2\,(g) + I_2\,(g) \leftrightarrow 2HI\,(g)$$

$$K = \frac{[HI]^2}{[H_2][I_2]}$$

Dissociation of Acids and Bases

Arrhenius definition of acids and bases

- **Acid** – a H-bearing compound that dissociates when dissolved in water
  - releasing H^+ ions (protons)
  - lowering pH
  - H^+ ions combine with water molecules to form hydronium ions $H_3O^+(aq)$.
- **Base** – a compound bearing the OH group that dissociates in water
  - releasing OH^- ions (hydroxide ions).
  - raising pH
  - OH^- combine with water to form hydrated ions
- According to the Law of Mass Action, for an acid (H_mA_n) and base (B_pOH_q) at equilibrium

$$H_mA_n \Leftrightarrow mH^+ + nA^-; \qquad K_a = \frac{[H^+]^m[A^-]^n}{[H_mA_n]}$$

$$B_pOH_q \Leftrightarrow pB^+ + qOH^-; \qquad K_b = \frac{[B^+]^p[OH^-]^q}{[B_pOH_q]}$$

- K_a and K_b are **dissociation constants**

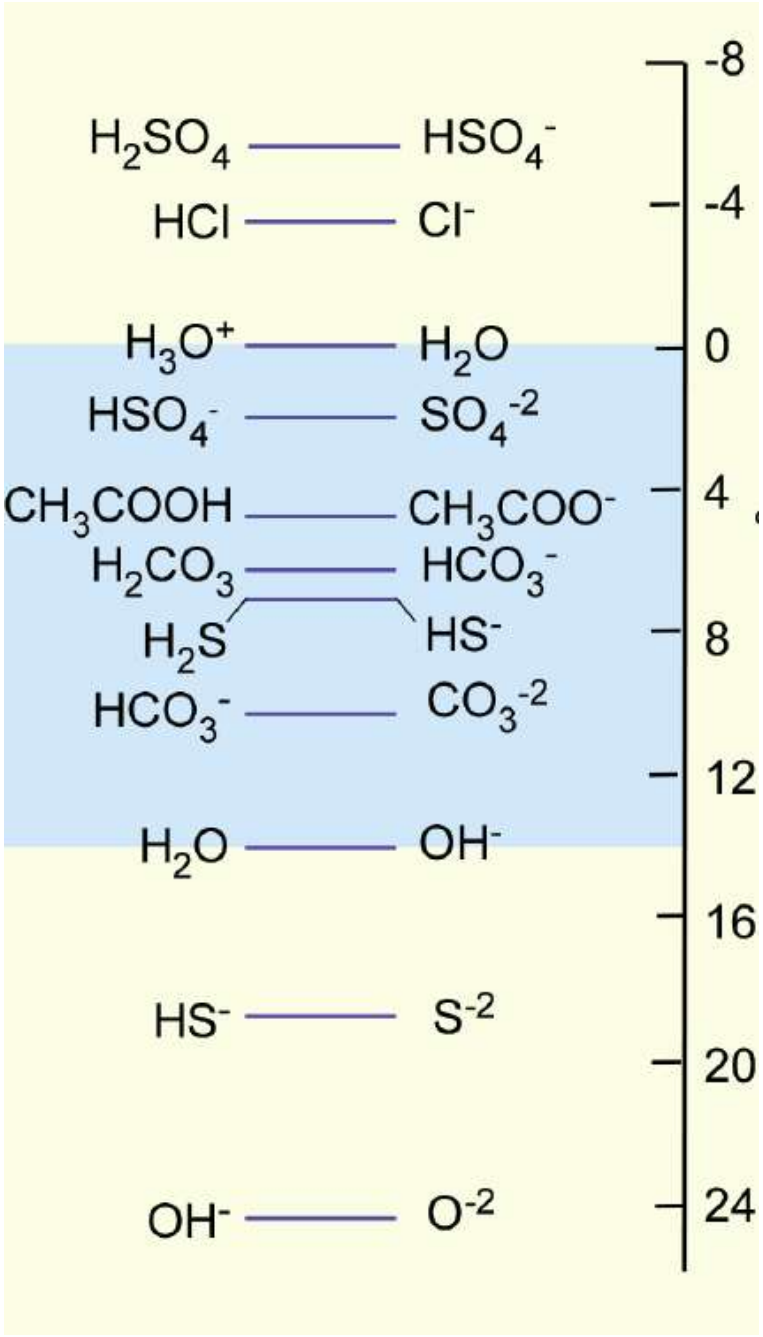
Strong and Weak Acids and Bases

- Strong Acids (e.g. HCl) dissociate almost completely in water.
- Weak acids (e.g. CH_3COOH – acetic acid) dissociate to a small extent ($K_a << 1$)
- Bases are also strong or weak depending on how much dissociation occurs when dissolved in water
- Do not confuse the **strength** of an acid or base (the extent to which it dissociates) with **concentration** (which is the amount dissolved in water).
- The degree of dissociation of an acid or base is given as:

$$\%D = \frac{\text{amount dissociated}}{\text{amount originally dissolved}} \times 100$$

- pH measures the acidity of a given solution (concentration and nature of the acid or base in solution)
- pK_a measures the strength (dissociation) of the acid (pK_b for bases)

pK_a and ladder diagrams



strong acids

Normal pH range

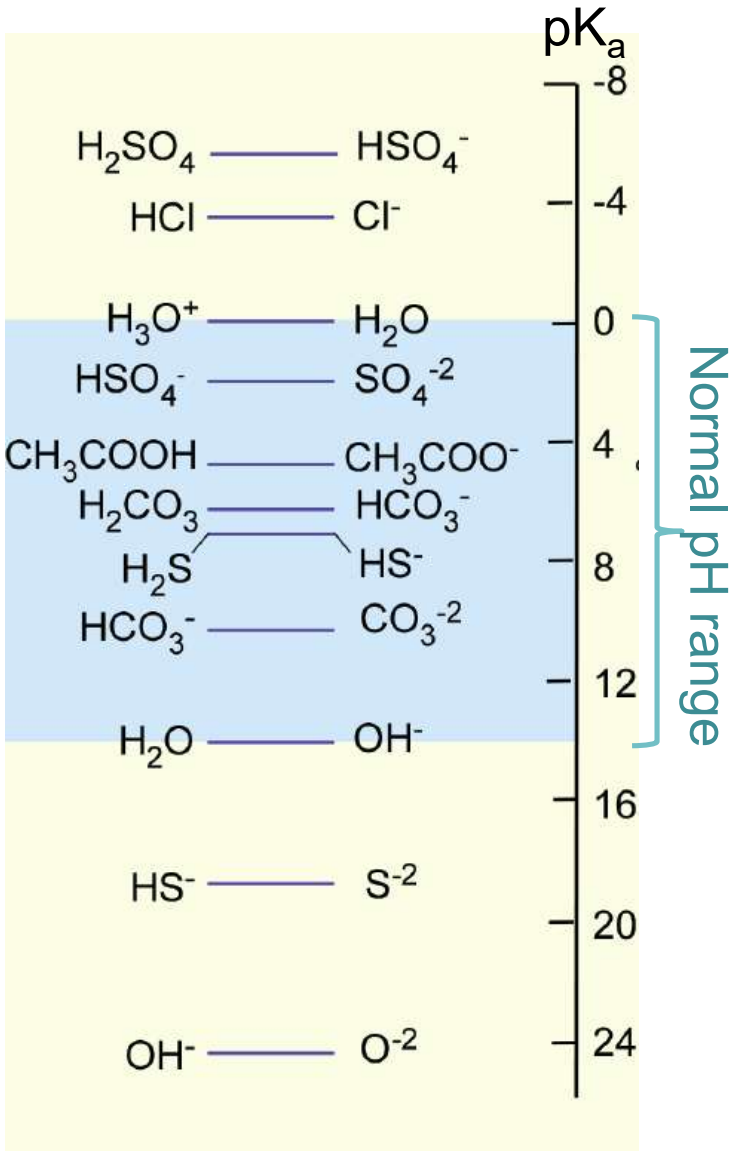
$$K_a = \frac{[H^+][A^-]}{[HA]} \quad \Rightarrow \quad pK_a = -\log K_a$$

$= 10$ then $-\log(10) = -1$

The higher K_a then the higher degree of dissociation = more acidic solution.

$K_a < 1$ then pK_a positive
$K_a = 1$ then $pK_a = 0$
$K_a > 1$ then pK_a negative

pK_a and pH



$$K_a = \frac{[H^+][A^-]}{[HA]} \quad \Rightarrow \quad pK_a = -\log K_a$$

$$[H^+] = K_a \frac{[HA]}{[A^-]}$$

$$pH = -\log[H^+] \; ; \; pK_a = -\log K_a \; ; \; -\log\left(\frac{A}{B}\right) = \log\left(\frac{B}{A}\right)$$

$$pH = pK_a + \log \frac{[A^-]}{[HA]}$$

Henderson-Hasselbalch equation

pKa helps you predict whether a molecule will dissociate at a specific pH

Example

Phenolphthalein is a pH indicator. It has pKa of 9.3 at room temperature. A- is pink. HA is colourless.
What is the ratio of A^- to HA and how does that explain the appearance of pink with increasing pH from 8.2 to 10?

pH 8.2:
$8.2 = 9.3 + \log (A^-/HA)$
$\log (A^-/HA) = -1.1$ therefore $A^-:HA = 10^{-1.1} = 0.08:1$

pH 10:
$10 = 9.3 + \log (A^-/HA)$
$\log (A^-/HA) = 0.7$ therefore $A^-:HA = 10^{0.7} = 5.01:1$

So at pH 8.2, molecules of A^- (pink) to HA (colourless) is 8:100
at pH 10, molecules of A^- (pink) to HA (colourless) is 501:100

Acid-Base properties of minerals

Most minerals react with water to raise the pH:

$2KAlSi_3O_8 + 2H_2O \rightarrow Al_2Si_2O_5(OH)_4 + 2K^+ + 2OH^-$

orthoclase *kaolinite*

$2Mg_2SiO_4 + 3H_2O \rightarrow Mg_3Si_2O_5(OH)_4 + Mg^{2+} + 2OH^-$

Olivine *Chrysotile*

$CaCO_3 + H_2O \rightarrow HCO_3^- + Ca^{2+} + OH^-$

A few minerals react with water to lower the pH:

$KFe_3(SO_4)_2(OH)_6 + 3H_2O \rightarrow K^+ + 3Fe(OH)_3 + 2SO_4^{2-} + 3H^+$

jarosite

Summary - 1

- The polarity of water results from hydrogen bonding
- The hydrogen bonding gives water properties that are vital for life
- When ionic compounds dissolve in water they dissociate into cations and anions.
- The ability of the dissolved ions to interact with each other is diminished by the presence of other ions and by interactions with the solvent molecules (hydrogen bonding).
- Therefore the molar concentrations are corrected for by factors known as activity coefficients.
- Activity coefficients decrease with ionic charge and increasing ionic strength of the solution
- Ionic strength is a measure of the concentration of ions in a solution

Summary - 2

- The Law of Mass Action can be used to obtain the equilibrium constant during dissolution.
- The equilibrium constant simply expresses the ratio of concentrations (or activities) of products/reactants for a chemical reaction at equilibrium.
- Acid and bases are compounds that dissociate in water to produce H^+ and OH^- ions respectively.
- The equilibrium constant for acids and bases is known as the dissociation constant K_a.
- The pH of a solution and Ka of the acid or base are related by the Henderson-Hasselbalch equation
- Most minerals are the salts of a strong acid and weak base.

Reading – Chapter 7 - Misra